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REVIEW

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Rational engineering of combinatorial bacterial therapies for cancer

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Abstract

Engineered bacteria are emerging as a transformative class of cancer therapeutics. Recent advances in synthetic biology have expanded the genetic circuit toolbox, enabling the programmable control of attenuation, payload release, and immunomodulation. These developments have transformed bacteria from simple, colonizing agents into a versatile chassis for complex therapeutic functions. In this review, we examine recent circuit-based strategies for enhancing tumor specificity, regulating therapeutic delivery and engaging the host immune system, with emphasis on programming spatiotemporal control and consortia behavior. We consider current barriers to clinical translational and discuss how rational engineering can guide the next generation of microbial therapeutics.

Keywords: Tumor microenvironment, Synthetic Biology, Cell therapy, Bacteria, CAR-T, Immunotherapy

Introduction

The convergence of synthetic biology and oncology has transformed bacteria from passive tumor colonizers into programmable living therapeutics. Although traditional cancer treatments like chemotherapy, radiotherapy, and immunotherapy, have dramatically improved patient outcomes, no single modality can fully address the biological and spatial complexity of solid tumors [1–3]. Irregular vasculature, metabolic heterogeneity, and necrotic regions limit the delivery and efficacy of conventional therapies, while systemic or repeated drug exposure can lead to dose-limiting toxicity in healthy tissues [4–6]. Bacterial species offer an alternative therapeutic option uniquely suited to these challenges: they remodel tumor architecture, activate localized antitumor immune responses through innate immunogenicity, and selectively colonize hypoxic and necrotic tumor regions [7, 8]. These natural properties provide a foundation for engineering bacteria as disease-specific therapeutic platforms capable of precision delivery within the tumor microenvironment (TME).

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Using bacteria as cancer therapy dates back more than a century to William Coley's pioneering use of live microbes to provoke tumor regression [9]. Although this approach was ultimately abandoned due to inconsistent outcomes and safety concerns, recent decades have witnessed its scientific revival [10]. Contemporary genetic engineering has redefined this once empirical pipeline as a programmable platform, enabling precise control over attenuation, safety, and payload release through rational engineering approaches [7, 8]. The surge in publications and clinical trials since the early 2000 s underscores a decisive shift from proof-of-concept to translational application, narrowing the gap between preclinical innovation and clinical deployment [11].

Early bacterial therapies made extensive use of facultative or obligate anaerobes such as *Escherichia coli*, *Salmonella*, and *Clostridium*, leveraging their intrinsic tumor tropism [10, 12, 13]. Recent progress in synthetic biology has dramatically expanded the capabilities of these systems. Higher-order genetic circuits now incorporate quorum sensing [14, 15], logic gating [16], synchronized lysis [17], and environmental feedback [18, 19] to enable precisely timed, tumor-restricted function. These advances have expanded bacterial therapies from single-function payload carriers to multi-layered control systems capable of sensing environmental cues, integrating complex logic, and coordinating behavior across populations [17, 18]. Such programmable circuits not only improve safety and therapeutic precision but also enable bacteria to dynamically respond to the tumor microenvironment, allowing treatments to adapt in real time as tumor conditions change.

In this review, we synthesize recent progress in engineering bacteria as living therapeutics, as well as key synthetic biology innovations that hold strong translational potential, but have yet to be applied in oncology. We emphasize how advances in genetic circuit architecture, environmental sensing, and population-level coordination are redefining how bacterial therapies are designed and conceptualized. By bridging the gap between clinically tested strategies and cutting-edge synthetic tools, we aim to define how bacterial design principles developed in model systems can inform therapeutic engineering in physiological contexts.

The convergence of these innovations has transformed therapeutic development into an engineering discipline, incorporating multilayered control of timing, localization, and coordinated population-level dynamics of therapeutic delivery [17, 18, 20]. Increasingly, these systems are being designed through combinatorial strategies that integrate multiple inputs [19, 21], layered feedback loops [14, 16], and modular payloads [17, 18, 22] to achieve context-specific precision within complex TMEs. Beyond optimizing circuit performance, recent work has expanded toward immunological integration, coupling engineered bacterial activity to host immune modulation and remodeling [23, 24]. These multi-tiered frameworks enable bacteria not only to deliver therapeutics but also to reshape the tumor–immune interface through localized cytokine release [17, 25, 26], antigen presentation [17, 27], or coordinated T-cell recruitment [25, 27].

By uniting applied bacterial therapies with advances from broader synthetic biology, we aim to hypothesize how higher-order genetic design can be translated into clinically relevant cancer therapeutics. Together these innovations point toward a new generation of bacterial therapies characterized by spatiotemporal precision, coordinated population

behavior, and immune orchestration, bridging the divide between preclinical ingenuity and practical implementation.

Spatiotemporal control systems

Translating early-stage synthetic biologics from bench to bedside requires improving the precision, stability, and control of therapeutic release [28]. Put more simply, one of the most important challenges in developing synthetic bacterial therapeutics, is not only what they can do, but when, where, and for how long they do it. Classic translational barriers, including evolutionary pressures in vivo, environmental heterogeneity, or variable growth dynamics, often prevent synthetic circuits that perform robustly in controlled environments from maintaining function in physiologically relevant contexts [29].

Recent advances in improving genetic stability, population-level coordination, environmental adaptation, and exogenously triggered actuation offer an unprecedented level of spatiotemporal control to address these challenges [11, 30] (Fig. 1). While many of these systems were originally developed outside of an oncological context, the underlying design principles of predictability, durability, and precision can be readily adapted to the tumor microenvironment, enabling highly specific, tumor-restricted activity.

Circuit stability and evolutionary robustness

Genetic instability remains one of the most significant barriers when bridging the translational gap in engineered cellular therapeutics [29]. In general, the more complex a circuit, the higher the metabolic burden it imposes [31, 32]. Without selective advantages such as antibiotic resistance, cells are prone to mutations, plasmid loss, or silencing

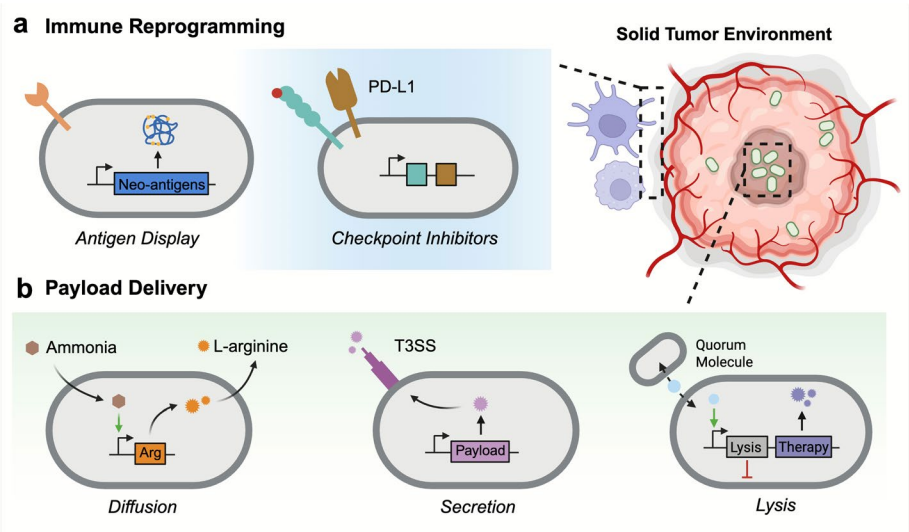


Fig. 1 Engineered bacterial strategies for immune modulation and therapeutic delivery in solid tumors. **a** *Immune reprogramming.* Bacteria can be engineered to display neoantigens or secrete checkpoint-modulating proteins, enabling targeted engagement of immune cells within the tumor microenvironment. **b** *Payload delivery.* Therapeutic molecules can be delivered through diffusion of metabolic or small-molecule payloads, active secretion systems such as T3SS, or quorum-regulated lysis circuits that coordinate population-wide release of intracellular therapies. Generated using Biorender

over time [29, 33]. To mitigate these issues, recent studies have developed systems that directly regulate replication dynamics, stabilize plasmid inheritance, and reduce mutational escape, maintaining functional expression over prolonged timescales.

Traditionally, multi-plasmid systems use distinct origins of replication to prevent incompatibility [34]. However, several recent studies have challenged this long-standing assumption, incorporating duplicate-origin systems in which different functional outputs are engineered into plasmids sharing the same replication origin (Fig. 2e). Instead

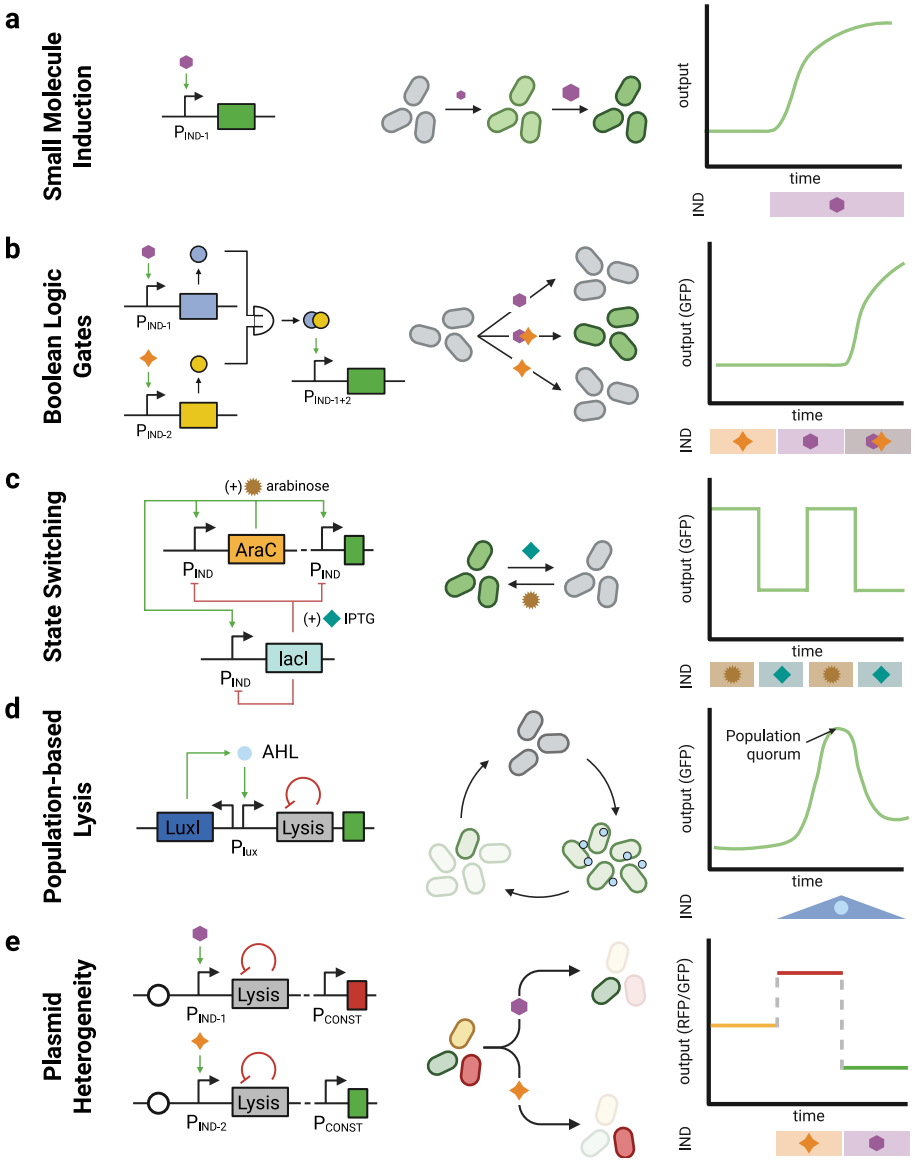


Fig. 2 Genetic Toolkit for inducible and population-based control in engineered bacterial therapeutics. **a** Small-molecule–induced circuits linking environmental inputs to reporter or lysis outputs. **b** Boolean logic gates enabling integration of multiple inducers to achieve conditional therapeutic activation. **c** Recombinase- and feedback-based state switching for temporal or memory-dependent control. **d** Population-based quorum-sensing lysis circuits coordinating synchronized release of intracellular payloads. **e** Plasmid coupled origin of replications for generating population-level variability and graded response. Generated using Biorender

of causing plasmid loss, this design creates a buffering capacity that minimizes metabolic burden while maintaining multiple outputs and tuning their relative expression according to environmental stimuli [35]. In practice, such systems enable precise phenotypic patterning within a single bacterial population, dynamically dividing it into two subpopulations with distinct functional outputs [36]. Mechanistically, this buffering protects against mutational inactivation, allowing cells to maintain circuit function significantly longer under evolutionary pressure [37]. Similarly, synthetic origins of replication can be designed to fine-tune replication rates and metabolic load, sustaining stable gene expression [38].

These buffering and replication-control strategies enable stable therapeutic output during extended intratumoral residence, where environmental pressures can rapidly erode engineered function. Functional "phenotypic shifting" has also been achieved through site-specific recombinases, allowing a single strain to execute distinct therapeutic regimes at different stages of tumor colonization—creating dynamic, adaptive treatments that counter tumor drug-resistance mechanisms [39]. Collectively, these systems demonstrate that stability need not constrain complexity; with proper control, even highly intricate genetic programs can maintain robust performance in vivo.

Engineered circuits must also maintain stability in population-wide coordination and buffer host interference in diffusible-signal exchange. While this can be traditionally achieved at the bench through antibiotic markers, alternative approaches must be used in physiological contexts. Several plasmid-borne strategies such as toxin-antitoxin systems [40, 41] and active partitioning modules [42] can encourage plasmid retention and regulate copy-number fluctuations. Auxotrophies specific to tumor environments [43] and essential gene manipulation [44] can confine plasmid retention in engineered strains to the TME. Orthogonal bacteria-specific promoters and RBS control mechanisms can insulate circuits from host interference [30]. Nonetheless, prolonged selective pressure specific to the tumor microenvironment can still drive mutational escape, underscoring the need for stabilization mechanisms tailored to therapeutic chassis and tumor-relevant conditions.

Population-level coordination

Population-level coordination offers an alternative approach to temporal control, synchronizing gene expression and therapeutic release across bacterial populations. Such collective behavior reduces single-cell noise, regulates dosage, and ensures coherent therapeutic activity [45].

One of the most influential examples is the synchronized lysis circuit (SLC), which couples quorum-sensing communication to collective lysis once a critical density is reached (Fig. 2d) [14]. This design enables pulsatile, rhythmic release of therapeutic payloads, as surviving cells reseed the population to regrow and repeat the cycle (Fig. 2e). The SLC framework has since been extended to deliver immunomodulatory payloads [8, 17], enhancing antitumor efficacy through controlled oscillations in therapeutic delivery.

Quorum-based coordination has also been integrated with TME niches to target specific tumor regions before payload release. For instance, obligate anaerobes that grow exclusively in hypoxic regions can be engineered with quorum-regulated systems that activate only in low-oxygen niches after reaching critical population thresholds [15].

Exogenous induction further adds external control layers, enabling clinicians to modulate bacterial activity or therapeutic dosage in real time [46]. Together, these strategies lay the groundwork for adaptive, feedback-informed dosing regimes guided by both microbial dynamics and patient response.

Tumor microenvironment-constrained drug release

Analogous to obligate anaerobe confinement in hypoxic regions, bacteria can be engineered to exploit specific chemical and metabolic niches of the TME, spatially confining therapeutic activity while minimizing off-target effects.

Recent delivery strategies have expanded on principles of tumor niche exploitation, leveraging metabolic gradients [7], hypoxia and necrosis [19], and remodeling of the extracellular matrix [47–49]. For example, bacterial surfaces have been functionalized with acid-responsive nanozymes that trigger payload release under low pH rather than relying on bacterial lysis, enabling full tumor colonization while achieving localized therapy release [50]. The intrinsic tumor-tropism of certain bacterial strains has also been harnessed to deliver immunomodulators [51] and CAR-T–supporting signals [27] directly within the TME, minimizing systemic exposure. Adaptive laboratory evolution (ALE) has further improved strain fitness and persistence under TME conditions, enhancing colonization and payload delivery without increasing circuit complexity [52].

An orthogonal approach involves exogenous control mechanisms, where external stimuli trigger or modulate bacterial activity post-administration. Examples include ultrasound-responsive gene expression [53], photothermal activation [54, 55], and magnetically guided control [56, 57]. Integrating such systems yields all-in-one theranostic designs in which diagnosis and treatment are coupled, such as photothermally active strains restricted to positively cancerous regions [58].

Collectively, these developments highlight spatiotemporal control as a critical frontier in bacterial therapeutic design. They also illustrate how technologies initially developed outside oncology can be adapted to overcome clinical translation barriers when integrated with cancer-focused engineering constraints.

Combinatorial approaches

Single-strain bacterial therapeutics have demonstrated remarkable potential to optimize tumor targeting, modulate the TME, deliver therapeutic payloads, and orchestrate localized immune responses [7, 8, 10]. However, solid tumors are heterogeneous, spatially complex niches [59, 60] that often demand equally complex interventions. Effective treatment frequently requires multi-modal strategies—combining therapeutic modalities in parallel [61], sequencing interventions with precise timing [62], or leveraging synergistic drug interactions with host immunity [63].

Synthetic biology has expanded our ability to rationally design systems that mimic these combinatorial paradigms (Fig. 2). Through multi-strain coordination [17, 18, 64], dual-payload release [17, 49, 55, 65], and hybrid-modality integration [21, 27, 66], researchers are now engineering bacterial therapies capable of synchronized, synergistic anti-tumor activity that surpasses the additive effects of individual interventions.

Engineered cellular consortia

A straightforward yet powerful strategy to overcome single-strain limitations is to combine strains into engineered consortia. Multi-strain treatments enable multiplexed drug delivery without compromising performance for any single payload [18]. By distributing functional burden across strains, these systems reduce per-strain genetic complexity while allowing multi-step interventions that would otherwise impose prohibitive metabolic costs. Inter-strain coordination can be maintained through quorum-sensing–based communication modules (Fig. 2d). The synchronized lysis circuit (SLC) exemplifies this principle [14]: bacteria monitor population density and synchronously lyse once a threshold is reached, releasing intracellular payloads into the interstitial space (Fig. 2d). This modular circuit can be readily adapted to deliver diverse payloads, and multi-strain SLC consortia releasing combinations of granulocyte-macrophage colony-stimulating factor (GM-CSF), α-PDL1, α-CTLA-4 nanobodies, chemokines, and cytokines have produced robust anti-tumor effects (Fig. 3a) [17, 65, 67].

Individual bacterial strains are known to stratify across physiological environments, forming distinct communities within niches like hypoxic cores, fibrous edges, and peri-tumoral regions, suggesting a coupled relationship between microbial fitness and local tumor architecture [68]. For example, engineered hypoxia-responsive strains and obligate anaerobes are likely to be most effective in poorly-vascularized necrotic regions, whereas mucosa-associated or commensal-derived chassis may prefer regions that retain higher oxygen and nutrient availability [69]. In this way, niche-adapted commensal chassis studies, in combination with our growing knowledge of the innate microbiome, provide a conceptual foundation for tailoring strain selection

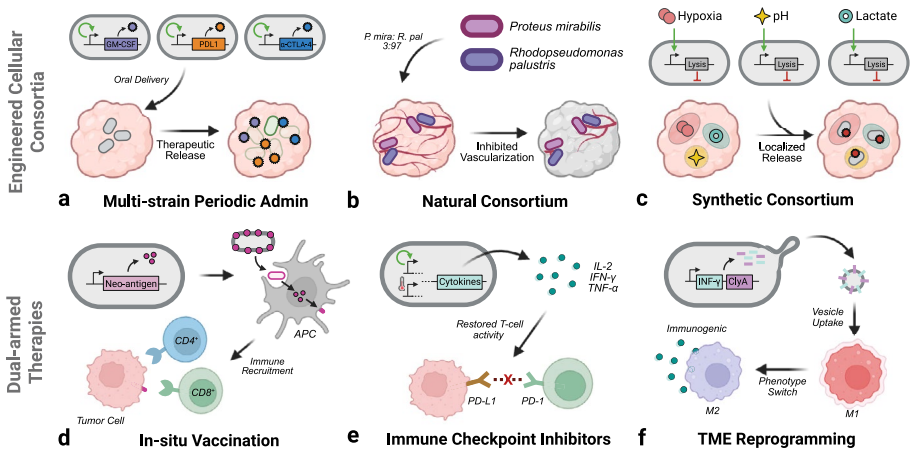


Fig. 3 Engineered consortia and dual-armed therapy frameworks for engineered bacterial therapeutics. **a** Multi-strain SLC administered periodically with three strains carrying immune-checkpoint payloads: GM-CSF, α-PDL1, and α-CTLA-4. **b** Administration of native TME bacterial consortia in defined proportions to inhibit tumor vasculature. **c** Multi-strain synthetic consortium programmed to release therapy under hypoxic, acidic (pH), or lactate-rich TME conditions. **d** In situ vaccination via bacterial production of neoantigens, APC uptake of bacteria, and presentation of tumor antigens. **e** Bacterial production of immune-modulatory factors (e.g., IL-2, IFN-γ, TNF-α) to recruit and activate T cells. **f** Bacterial vesicle–mediated delivery of IFN-γ and ClyA for reprogramming macrophages from an inflammatory M1 phenotype to an immunosuppressive M2 state and broader TME remodeling. Generated using Biorender

and genetic constructs to tumor-specific archetypes, eventually resolving consortia designs that partition strains and treatments by tumor sub-region.

Population composition within a consortium can also serve as a regulatory parameter, introducing an ecological layer of control that enhances fine-tuning of therapeutic output. Natural inter-strain competition can be leveraged to dynamically balance complementary populations, maintaining optimal ratios as environmental conditions evolve (Fig. 3b) [64]. In engineered systems, these dynamics can be harnessed to remodel the TME in response to disease progression, sustain anti-tumor responses, and prevent resistance. Quorum-based feedback and density-dependent promoters further stabilize population ratios and allow therapies to adapt dynamically to tumor evolution.

Beyond cooperation, division of labor within bacterial consortia enables distinct strains to perform specialized and complementary functions. For example, the synthetic consortium SynCon comprises individual strains programmed to release chemotherapeutics in response to hypoxia, acidity, or elevated lactate, collectively achieving spatially comprehensive treatment across the TME (Fig. 3c) [18]. Division of labor can also extend across modalities. For instance, bacterial strains can serve as tumor-homing guides for engineered CAR-T or CAR-NK cells [22, 27], improving immune localization and reducing off-target toxicity. By enhancing immune-cell infiltration and mitigating functional exhaustion, bacterial consortia expand the therapeutic reach of adoptive cell therapies into solid tumor contexts.

Dual-armed therapies

Beyond engineered consortia, synthetic bacteria can directly interface with host immunity to simultaneously elicit multiple anti-tumor responses [23–25]. These interactions reprogram the immune landscape while complementing traditional therapies, effectively transforming bacteria into dynamic in situ adjuvants that stimulate inflammation and promote systemic anti-tumor immunity.

One approach involves the coordinated delivery of tumor-associated antigens (TAAs) or neoantigens (Fig. 3d). This strategy, often described as in situ vaccination, drives T-cell priming by local antigen-presenting cells (APCs) and has shown potent anti-tumor efficacy across multiple models [16]. The tumor-homing ability of engineered bacteria, coupled with TME-regulated gene circuits, makes them ideal vehicles for localized antigen delivery.

Synthetic bacteria can also be designed to harness or potentiate immune checkpoint blockade (ICB) regimes, a therapy often limited by systemic toxicity and poor tumor penetration [70] (Fig. 3e). Engineered strains have been constructed to directly produce ICB molecules [65] and to regulate their local delivery through dual-input logic circuits that restrict pro-inflammatory cytokine (e.g., IFN-γ) expression to tumor regions, thereby amplifying local immune responses while minimizing systemic effects [25]. Temperature-responsive promoters have further enabled coupling of immunotherapy with photothermal activation for precise spatial control [26].

Engineered bacteria also modulate the extracellular matrix (ECM) to enhance drug perfusion. ECM remodeling has been achieved through enzymatic degradation [47, 48] or inhibition of matrix synthesis [23], improving penetration of ICBs such as α-PDL1. Complementary strategies, such as hyperbaric oxygen therapy, can further deplete the

dense ECM and synergize with facultative anaerobic bacteria to enhance tumor penetration and potentiate phototherapies and ICB responses [21].

Tumor-associated macrophages (TAMs) offer another alternative for immune reprogramming (Fig. 3f). While bacterial components such as lipopolysaccharide (LPS) naturally engage macrophages [71], bacteria have also been engineered to actively polarize macrophages toward the pro-inflammatory M1 phenotype via attenuation [72] or cytokine delivery [24]. Some systems further employ bacterial vectors to deliver oncolytic viral components, creating coordinated bacterial–viral therapies that enhance tumor killing through orthogonal mechanisms [20].

Combinatorial approaches such as these represent a major evolutionary step in the field of bacterial cancer therapy. By distributing genetic and functional complexity across coordinated systems, researchers can integrate multiple payloads, signaling networks, and therapeutic modalities while maintaining metabolic feasibility and control. Communication modules, population-level feedback, and sequential circuit architectures together define a new frontier for precise, scalable, and adaptive bacterial therapeutics.

Outlook

Expanding circuit modularity and compatibility

As genetic circuits grow in complexity, it becomes increasingly difficult to extend designs across bacterial species and physiological contexts. A persistent challenge is achieving portability, ensuring that constructs remain functional when transferred between species or environmental conditions. Most synthetic systems have been built in a limited set of laboratory chassis such as *E. coli* and *Salmonella Typhimurium*. While these organisms offer established safety profiles and engineering toolkits, they are not always optimal for tumor colonization, immune modulation, or long-term persistence. Expanding synthetic biology into new hosts will therefore require frameworks that standardize circuit architecture and minimize the need for re-engineering across species.

To this end, several systems have been developed to modularize circuit design and achieve cross-species functionality. The Synthetic Modularity for Advanced Rational Therapeutics (SMARTs) framework employs a standardized architecture of promoters, ribosome binding sites (RBSs), and insulator sequences that maintain consistent performance across both Gram-negative and Gram-positive species [73]. SMARTs enables rapid substitution of sensing and response modules, allowing rational tuning of circuit behavior across diverse hosts without extensive redesign. When applied within microbial consortia, SMARTs could enable coordinated responses across not only multiple strains but multiple species, creating higher-order therapeutic micro-ecologies.

A complementary approach, the Universal Bacterial Expression Resource (UBER) system, achieves cross-species functionality by decoupling transcriptional control from host-specific regulatory networks [74]. By leveraging orthogonal sigma factors and promoter sets, UBER circuits function independently of native transcriptional machinery, ensuring predictable operation even in non-model bacterial hosts. Together, SMARTs and UBER establish a foundation for universal, modular circuit design, supporting "plug-and-play" adaptability, high-throughput prototyping, and standardized performance benchmarking.

In an oncological context, chassis selection is guided not by ease of manipulation but by tumor tropism, immunogenicity, and safety [75]. A universal circuit framework would thus enable translation of sophisticated designs into clinically viable species optimized for therapeutic delivery, bridging the gap between bench-ready constructs and patient-ready medicines.

Bridging the translational gap

Despite remarkable progress in circuit complexity, most bacterial therapies that have advanced to clinical testing rely on minimal genetic modifications [76, 77] or single-payload expression [78, 79], while higher-order systems remain preclinical. Clinical efforts have also centered largely on a narrow set of chassis with established safety profiles, such as E. coli Nissle 1917, while commercially available engineered bacteria are limited to non-therapeutic products such as Bacillus subtilis in the consumer microbiome sector [80], which fall outside medical regulatory frameworks. To date, the FDA has yet to progress an engineered strain through clinical trials into commercial availability for the treatment of cancer. This gap reflects both technical uncertainty and regulatory conservatism, as achieving stability, predictable behavior, and reliable containment in vivo remains significantly more challenging for complex systems.

Future translation will depend on demonstrating that advanced architectures can maintain reliability in vivo and be manufactured at clinically relevant scales. One promising strategy is a modular validation framework, in which each genetic component is independently assessed for safety, efficacy, and stability before integration into therapeutic strains. This type of regulatory framework would operate much in the same way a parts registry does for device engineering. While full-strain testing remains essential, component-level validation enables standardized benchmarking, de-risks circuit assembly, and simplifies regulatory review by establishing reusable data on well-defined modules within an in vivo context. In parallel, component interactions can be evaluated in vitro prior to integration in final therapeutic strains, streamlining and thereby expediting end-product development. Such an approach would accelerate regulatory approval while maintaining rigorous performance validation throughout the design–build–test cycle.

Furthermore, bacterial colonization and therapeutic performance vary substantially across disease stages, and different engineering strategies align more naturally with specific tumor contexts [7]. Hypoxia-targeted strains and quorum-based activation circuits are generally best suited for early or localized solid tumors where spatial heterogeneity and necrotic niches support robust intratumoral enrichment [7, 81], whereas engineered consortia and systemically delivered strains may offer greater utility in metastatic or disseminated disease due to their broader tissue reach. These are important considerations when designing clinically relevant constructs, and are crucial in maintaining therapeutic efficacy and strain performance while minimizing circuit complexity.

Balancing complexity with practicality

Even as modular systems mature, clinical feasibility remains a central hurdle for implementing higher-order circuitry. Therapeutic constructs must not only showcase innovation but also remain tractable, safe, and scalable. Thus, pragmatism in circuit design

must be prioritized. Each added layer of regulation should provide clear functional benefit without imposing excessive metabolic or regulatory overhead.

Consortia-based and division-of-labor strategies provide a practical solution to this balance. Modularizing genetic functions across multiple strains reduces per-strain complexity while preserving collective sophistication. Early examples have demonstrated second- and third-order execution of distributed functions, establishing a precedent for expanding into higher-dimensional therapeutic systems. By distributing metabolic burden or employing dynamic population control, engineered bacteria can generate spatiotemporally coordinated therapeutic "cocktails" in situ within tumors. Coupled with the strain diversification enabled by frameworks such as SMARTs and UBER, these strategies represent a paradigm shift toward "plug-and-play" bacterial therapeutics, simplifying manufacturing and regulation while opening new opportunities in precision and personalized medicine.

Towards personalized and precision medicine

Universal and modular circuit frameworks also provide a path toward tailored bacterial therapeutics, enabling constructs to be matched to specific tumor environments, molecular profiles, and patient needs. Therapeutic programs could be customized to the biochemical and spatial features of an individual's tumor microenvironment.

In practice, this might involve selecting from a library of validated payloads, matching their expression modules to characterized control circuits, and assembling them into a multi-strain system optimized for a given tumor profile. Consortia compositions could then be fine-tuned in real time based on disease state or clinical response. As omics-based profiling becomes routine, these personalized constructs could be paired with tumor-specific signatures in much the same way targeted small-molecule therapies are today.

The success of such a vision will depend on the convergence of universal design frameworks, modular validation, precision chassis selection, and standardized regulatory benchmarks, combined with the ongoing incorporation of mechanistic insights from both synthetic and tumor biology. Collectively, these advances position bacterial engineering to evolve from a proof-of-concept frontier into a standardized, clinically translatable discipline, ushering in programmable living medicines for precision oncology.

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Data availability
No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare the following competing financial interest(s): J.H. declares that he is a co-founder of GenCirq Inc, which focus on cancer therapeutics. He is on the Board of Directors and has equity in GenCirq. His spouse is employed part time by GenCirq for bookkeeping and employee support with Human Resources.

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